

## KCET-2016-Set-A-1

### Questions – Physics

1. The body falls freely for 10 s. Its average velocity for this journey is (Take,  $g = 10 \text{ ms}^{-2}$ )
  - (1)  $100 \text{ ms}^{-1}$
  - (2)  $10 \text{ ms}^{-1}$
  - (3)  $50 \text{ ms}^{-1}$
  - (4)  $5 \text{ ms}^{-1}$
  
2. Three projectiles A, B and C are projected at an angle of  $30^\circ$ ,  $45^\circ$  and  $60^\circ$  respectively. If  $R_A$ ,  $R_B$  and  $R_C$  are the range of A, B and C respectively then, (Velocity of projection is same for A, B and C)
  - (1)  $R_A = R_B = R_C$
  - (2)  $R_A = R_C > R_B$
  - (3)  $R_A < R_B < R_C$
  - (4)  $R_A = R_C < R_B$

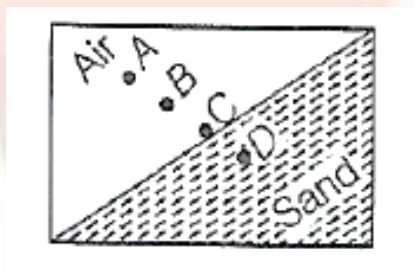
3. The component of a vector  $r$  along X-axis will have a maximum value, if
- (1)  $r$  is along positive X-axis.
  - (2)  $r$  is along positive Y-axis.
  - (3)  $r$  is along negative Y-axis.
  - (4)  $r$  makes an angle of  $45^\circ$  with the X-axis.
4. The maximum acceleration in a train in which a 50 kg box lying on its floor will remain stationary (Given coefficient of static friction between the box and the train's floor 0.3 and  $g = 10 \text{ ms}^{-2}$ )
- (1)  $5.0 \text{ ms}^{-2}$
  - (2)  $3.0 \text{ ms}^{-2}$
  - (3)  $1.5 \text{ ms}^{-2}$
  - (4)  $15 \text{ ms}^{-2}$
5. A 12 kg bomb at rest explodes into two pieces of 4 kg and 8 kg. If momentum of 4 kg piece is 20 Ns, the kinetic energy of 8 kg piece is
- (1) 25 J

(2) 20 J

(3) 50 J

(4) 40 J

6. Which of the point is likely position of the center of mass of the system shown in figure?



(1) A

(2) D

(3) B

(4) C

7. Three bodies a ring ( $R$ ), a solid cylinder ( $C$ ) and a solid sphere ( $S$ ) having same mass and same radius roll down the inclined plane without slipping. They start from rest, if  $v_R$ ,  $v_C$  and  $v_S$  are velocities of respective bodies on reaching the bottom of the plane, then

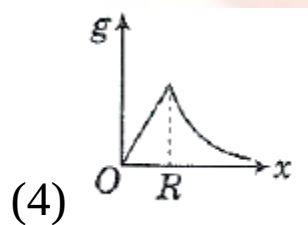
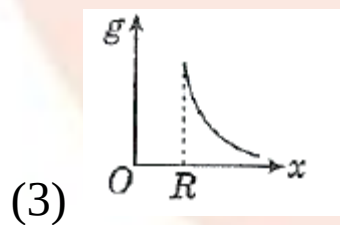
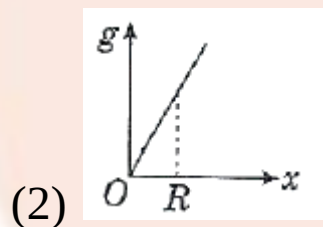
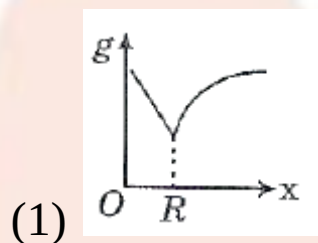
(1)  $v_R = v_C = v_S$

$$(2) v_R > v_C > v_S$$

$$(3) v_R < v_C < v_S$$

$$(4) v_R = v_C > v_S$$

8. Variation of acceleration due to gravity ( $g$ ), with distance  $x$  from the center of Earth is best represented by ( $R \rightarrow$  Radius of the Earth)



9. A spring is stretched by applying a load at its free end. The strain produced in the spring is

- (1) volumetric
- (2) shear
- (3) longitudinal and shear
- (4) longitudinal

10. An ideal fluid flows through a pipe of circular cross-section with diameters 5 cm and 10 cm as shown in the figure. The ratio of velocities of fluid at A and B is



- (1) 4:1
- (2) 1:4
- (3) 2:1
- (4) 1:2

11. A pan filled with hot food cools from  $94^{\circ}\text{C}$  to  $86^{\circ}\text{C}$  in 2 min. When the room temperature is  $20^{\circ}\text{C}$  How long will it cool from  $74^{\circ}\text{C}$  to  $66^{\circ}\text{C}$ ?

- (1) 2 min

(2) 2.8 min

(3) 2.5 min

(4) 1.8 min

12. Four rods of same material with different radii  $r$  and length  $l$  are used to connect two heat reservoirs at different temperature.

Which one will conduct most heat?

(1)  $r = 1 \text{ cm}$ ,  $l = 1 \text{ m}$

(2)  $r = 1 \text{ cm}$ ,  $l = \frac{1}{2} \text{ m}$

(3)  $r = 2 \text{ cm}$ ,  $l = 2 \text{ m}$

(4)  $r = 2 \text{ cm}$ ,  $l = \frac{1}{2} \text{ m}$

13. A Carnot engine working between 300 K and 400 K has 800 J of useful work. The amount of heat energy supplied to the engine from the source is

(1) 2400 J

(2) 3200 J

(3) 1200 J

(4) 3600 J

14. A particle executing SHM has a maximum speed of  $0.5 \text{ ms}^{-1}$  and maximum acceleration of  $1.0 \text{ ms}^{-2}$ . The angular frequency of acceleration is

(1)  $2 \text{ rad s}^{-1}$

(2)  $0.5 \text{ rad s}^{-1}$

(3)  $2\pi \text{ rad s}^{-1}$

(4)  $0.5\pi \text{ rad s}^{-1}$

15. A source of sound is moving with a velocity of  $50 \text{ ms}^{-1}$  towards a stationary observer. The observer measures the frequency of sound as 500 Hz. The apparent frequency of sound as heard by the observer when source is moving away from him with the same speed is (Speed of sound at room temperature is  $350 \text{ ms}^{-1}$ )

(1) 400 Hz

(2) 666 Hz

(3) 375 Hz

(4) 177.5 Hz

16. If there are only one type of charge in the universe, then

( $E \rightarrow$  Electric field,  $dS \rightarrow$  Area vector)

(1)  $\oint E \cdot dS \neq 0$  on any surface

(2)  $\oint E \cdot dS$  could not be defined

(3)  $\oint E \cdot dS = \infty$  if charge is inside

(4)  $\oint E \cdot dS = 0$  if charge is outside,  $= \frac{q}{\epsilon_0}$  if charge is inside

17. An electron of mass  $m$ , charge  $e$  falls through a distance  $h$  metre in a uniform electric field  $E$ , then time of fall

(1)  $t = \sqrt{\frac{2hm}{eE}}$

(2)  $t = \frac{2hm}{eE}$

(3)  $t = \sqrt{\frac{2eE}{hm}}$

(4)  $t = \frac{2eE}{hm}$



18. If  $E_{ax}$  and  $E_{eq}$  represents electric field at a point on the axial and equatorial line of a dipole. If points are at a distance  $r$  from the centre of the dipole, for  $r \gg a$

(1)  $E_{ax} = E_{eq}$

(2)  $E_{ax} = -E_{eq}$

(3)  $E_{ax} = -2E_{eq}$

(4)  $E_{ax} = 2E_{eq}$

19. Nature of equipotential surface for a point charge is

(1) ellipsoid with charge at foci

(2) sphere with charge at the center of the sphere

(3) sphere with charge on the surface of the sphere

(4) plane with charge on the surface

20. A particle of mass 1 g and charge  $1 \mu\text{C}$  is at rest on a frictionless horizontal surface at distance 1 m from the fixed charge 2 mC. the particle is released, it will be repelled. The speed of the particle when it is at a distance of 10 m from the fixed charge

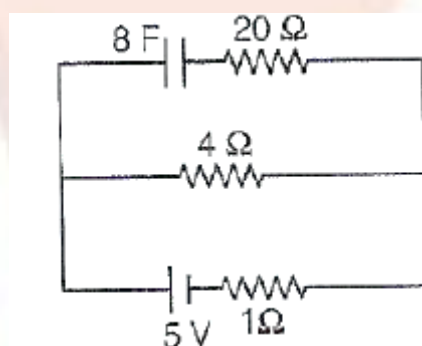
(1)  $60 \text{ ms}^{-1}$

(2)  $100 \text{ ms}^{-1}$

(3)  $90 \text{ ms}^{-1}$

(4)  $180 \text{ ms}^{-1}$

21. A capacitor of  $8 \text{ F}$  is connected as shown in the figure. Charge on the plates of the capacitor



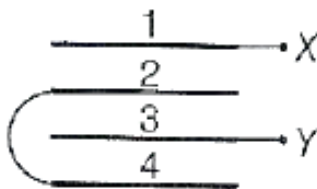
(1)  $32 \text{ C}$

(2)  $40 \text{ C}$

(3)  $0 \text{ C}$

(4)  $80 \text{ C}$

22. Four metal plates are arranged as shown in the figure. Capacitance between X and Y ( $A \rightarrow$  Area of each plate,  $d \rightarrow$  distance between the plates) is



(1)  $\frac{3 \epsilon_0 A}{2 d}$

(2)  $\frac{2 \epsilon_0 A}{d}$

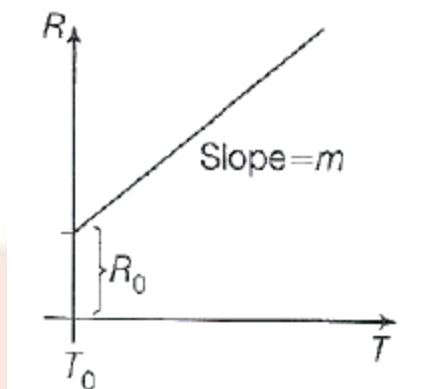
(3)  $\frac{2 \epsilon_0 A}{3 d}$

(4)  $\frac{3 \epsilon_0 A}{d}$

23. Mobility of free electrons in a conductor is

- (1) directly proportional to electron density
- (2) directly proportional to relaxation time
- (3) inversely proportional to electron density
- (4) inversely proportional to relaxation time

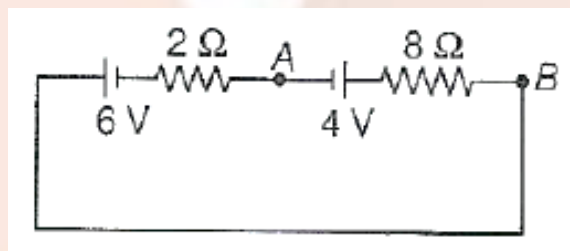
24. Variation of the resistance of conductor with temperature is as shown



The temperature coefficient ( $\alpha$ ) of the conductor is

- (1)  $\frac{R_0}{m}$
- (2)  $mR_0$
- (3)  $m^2 R_0$
- (4)  $\frac{m}{R_0}$

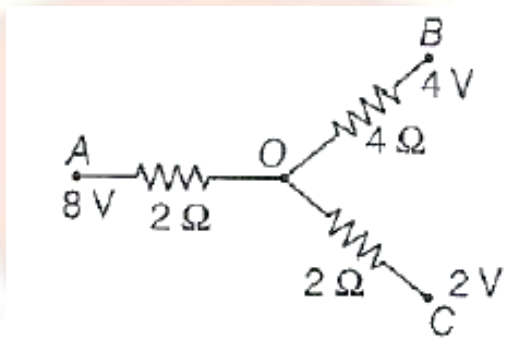
25. Potential difference between A and B in the following circuit



- (1) 4 V
- (2) 5.6 V
- (3) 2.8 V

(4) 6 V

26. In the following network, potential at O is



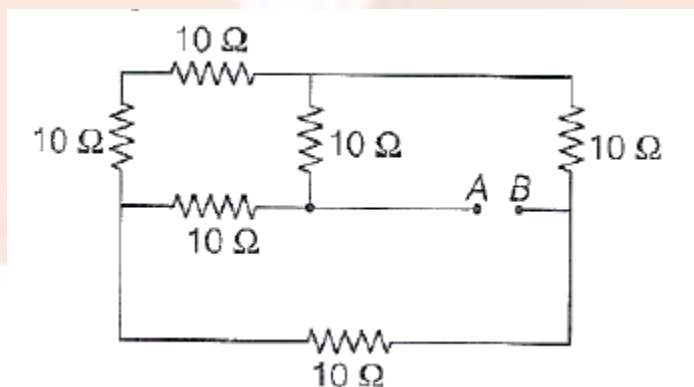
(1) 4 V

(2) 3 V

(3) 6 V

(4) 4.8 V

27. Effective resistance between A and B in the following circuit is



(1)  $10\ \Omega$

(2)  $20\ \Omega$

(3)  $5\ \Omega$

(4)  $\frac{20}{3}\ \Omega$

28. Two heating coils of resistances  $10\ \Omega$  and  $20\ \Omega$  are connected in parallel and connected to a battery of emf  $12\ \text{V}$  and internal resistance  $1\ \Omega$ . The power consumed by them are in the ratio

(1)  $1:4$

(2)  $1:3$

(3)  $2:1$

(4)  $4:1$

29. A proton is projected with a uniform velocity  $v$  along the axis of a current-carrying solenoid, then

(1) the proton will be accelerated along the x-axis

(2) the proton path will be circular about the axis

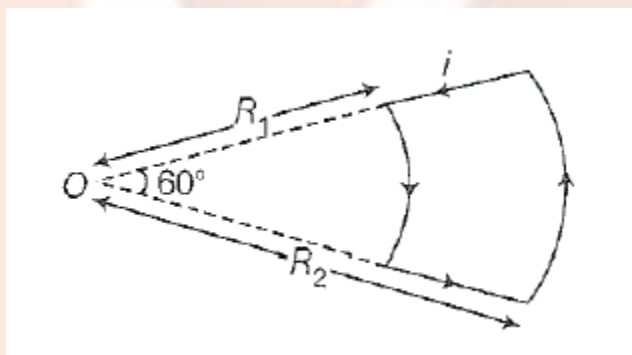
(3) the proton move along helical path

(4) the proton will continue to move with velocity  $v$  along the axis

30. In the cyclotron, as radius of the circular path of the charged particle increases ( $\omega$  = angular velocity,  $v$  = linear velocity)

- (1) both  $\omega$  and  $v$  increases
- (2)  $\omega$  only increases,  $v$  remains constant
- (3)  $v$  increases,  $\omega$  remains constant
- (4)  $v$  increases,  $\omega$  decreases

31. A conducting wire carrying current is arranged as shown in the figure. The magnetic field at O is



- (1)  $\frac{\mu_0 i}{12} \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]$
- (2)  $\frac{\mu_0 i}{12} \left[ \frac{1}{R_1} + \frac{1}{R_2} \right]$
- (3)  $\frac{\mu_0 i}{6} \left[ \frac{1}{R_1} - \frac{1}{R_2} \right]$

$$(4) \frac{\mu_0 i}{6} \left[ \frac{1}{R_1} + \frac{1}{R_2} \right]$$

32. The quantity of a charge that will be transferred by a current flow of 20 A over 1 h 30 min period is

- (1)  $10.8 \times 10^3 \text{ C}$
- (2)  $10.8 \times 10^4 \text{ C}$
- (3)  $5.4 \times 10^3 \text{ C}$
- (4)  $1.8 \times 10^4 \text{ C}$

33. A galvanometer coil has a resistance of  $50 \Omega$  and the meter shows full scale deflection for a current of 5 mA. This galvanometer is converted into voltmeter of range 0-20 V by connecting

- (1)  $3950 \Omega$  in series with Galvanometer
- (2)  $4050 \Omega$  in series with Galvanometer
- (3)  $3950 \Omega$  in parallel with Galvanometer
- (4)  $4050 \Omega$  in parallel with Galvanometer



34.  $\chi_1$  and  $\chi_2$  are susceptibility of a paramagnetic material at temperature  $T_1$  K and  $T_2$  K respectively, then

(1)  $\chi_1 = \chi_2$

(2)  $\chi_1 T_1 = \chi_2 T_2$

(3)  $\chi_1 T_2 = \chi_2 T_1$

(4)  $\chi_1 \sqrt{T_1} = \chi_2 \sqrt{T_2}$

35. At certain place, the horizontal component of Earth's magnetic field is 3.0 G and the angle dip at that place is  $30^\circ$ . The magnetic field of Earth at that location is

(1) 4.5 G

(2) 5.1 G

(3) 3.5 G

(4) 6.0 G

36. The process of superimposing message signal on high frequency carrier wave is called

(1) amplification

(2) demodulation

(3) transmission

(4) modulation

37. A long solenoid with 40 turns per cm carries a current of 1 A.

The magnetic energy stored per unit volume is .....  $\text{Jm}^{-3}$

(1)  $3.2\pi$

(2)  $32\pi$

(3)  $1.6\pi$

(4)  $6.4\pi$

38. A wheel with 10 spokes each of length  $L$  m is rotated with a uniform angular velocity  $\omega$  in a plane normal to the magnetic field  $B$ . The emf induced between the axle and the rim of the wheel is

(1)  $\frac{1}{2}N\omega BL^2$

(2)  $\frac{1}{2}\omega BL^2$

(3)  $\omega BL^2$

(4)  $N\omega BL^2$

39. The rms value of the current is 50 Hz AC of circuit is 6 A. The average value of AC current over a cycle is

(1)  $6\sqrt{2}$

(2)  $\frac{3}{\pi\sqrt{2}}$

(3) zero

(4)  $\frac{6}{\pi\sqrt{2}}$

40. A capacitor of capacitance  $10\mu\text{F}$  is connected to an AC ammeter. If the source voltage varies as  $V = 50\sqrt{2}\sin 100t$ , the reading of the ammeter is

(1) 50 mA

(2) 70.7 mA

(3) 5.0 mA

(4) 7.07 mA

41. In a series LCR circuit, the potential drop across L, C and R respectively are 40 V, 120 V and 60 V. Then the source voltage is

- (1) 220 V
- (2) 160 V
- (3) 180 V
- (4) 100 V

42. In a series LCR circuit, an alternating emf ( $V$ ) and current ( $i$ ) are given by the equation  $V = V_0 \sin \omega t$ ,  $I = I_0 \sin\left(\omega t + \frac{\pi}{3}\right)$ . The average power dissipated in the circuit over a cycle of AC is

- (1)  $\frac{V_0 I_0}{2}$
- (2)  $\frac{V_0 I_0}{4}$
- (3)  $\frac{\sqrt{3}}{2} V_0 I_0$
- (4) zero

43. Electromagnetic radiation used to sterilize mil is

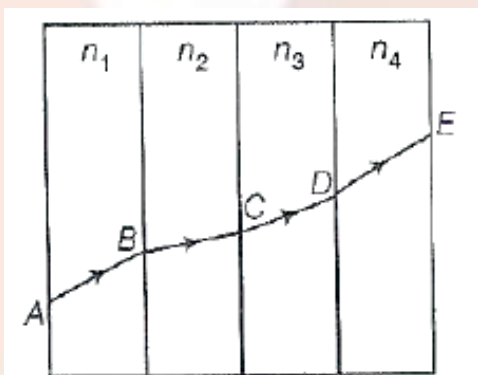
- (1) X-rays
- (2)  $\gamma$ -rays
- (3) UV rays

(4) Radio waves

44. A plane glass plate is placed over the various colored letters (violet, green, yellow, red) the letter which appear to raised more is

- (1) red
- (2) yellow
- (3) green
- (4) violet

45. A ray of light passes through four transparent media with refractive index  $n_1, n_2, n_3$  and  $n_4$  as shown in the figure. The surfaces of all media are parallel



If the emergent ray DE is parallel to incident ray AB, then

- (1)  $n_1 = n_4$

$$(2) \ n_2 = n_4$$

$$(3) \ n_3 = n_4$$

$$(4) \ n_1 = \frac{n_2 + n_3 + n_4}{3}$$

46. Focal length of a convex lens is 20 cm and its refractive index is 1.5. It produces an erect, enlarged image if the distance of the object from the lens is

(1) 40 cm

(2) 30 cm

(3) 50 cm

(4) 20 cm

47. A ray of light suffers a minimum deviation when incident on a equatorial prism of refractive index  $\sqrt{2}$ . The angle of deviation is

(1)  $30^\circ$

(2)  $45^\circ$

(3)  $60^\circ$

(4)  $50^\circ$

48. In Young's double slit Experiment, the source is white light. One slit is covered with red filter and other with blue filter. There shall be

- (1) alternate red and blue fringes
- (2) alternate dark and pink fringes
- (3) alternate dark and yellow fringes
- (4) no interference

49. Light of wavelength 600 nm is incident normally on a slit of width 0.2 mm. The angular width of central maxima in the diffraction pattern is (measured from minimum to minimum)

- (1)  $6 \times 10^{-3}$  rad
- (2)  $4 \times 10^{-3}$  rad
- (3)  $2.4 \times 10^{-3}$  rad
- (4)  $4.5 \times 10^{-3}$  rad

50. For what distance ray optics is good approximation when the aperture is 4 mm and the wavelength of the light is 400 nm?

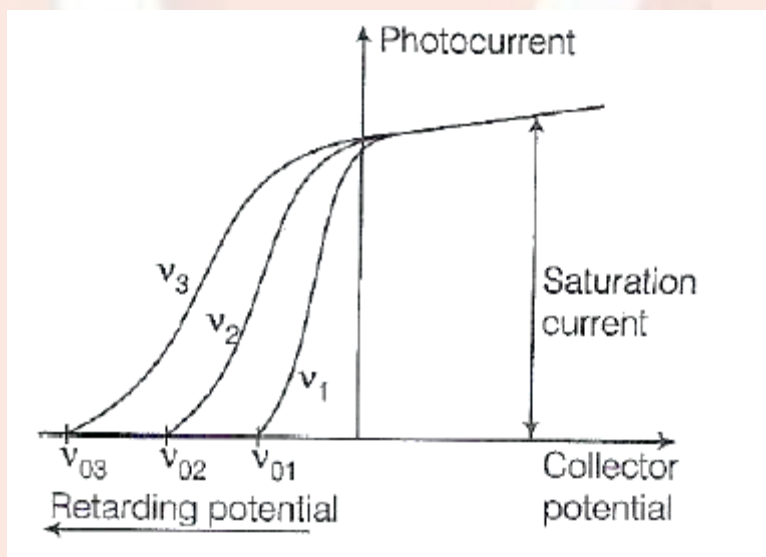
- (1) 24 m

(2) 40 m

(3) 18 m

(4) 30 m

51. The variation of photocurrent with collector potential for different frequencies of incident radiation  $\nu_1$ ,  $\nu_2$  and  $\nu_3$  is as shown in the graph, then



(1)  $\nu_1 = \nu_2 = \nu_3$

(2)  $\nu_1 > \nu_2 > \nu_3$

(3)  $\nu_1 < \nu_2 < \nu_3$

(4)  $\nu_3 = \frac{\nu_1 + \nu_2}{2}$



52. The de-Broglie wavelength of an electron accelerated to a potential of 400 V is approximately

- (1) 0.03 nm
- (2) 0.04 nm
- (3) 0.12 nm
- (4) 0.06 nm

53. Total energy of an electron in excited state of hydrogen atom is  $-3.4$  eV. The kinetic and potential energy of electron in this state are

- (1)  $K = -3.4$  eV,  $U = -6.8$  eV
- (2)  $K = 3.4$  eV,  $U = -6.8$  eV
- (3)  $K = -6.8$  eV,  $U = +3.4$  eV
- (4)  $K = +10.2$  eV,  $U = -13.6$  eV

54. When electron jumps from  $n = 4$  level to  $n = 1$  level, the angular momentum of electron changes by

- (1)  $\frac{h}{2\pi}$

(2)  $\frac{2h}{2\pi}$

(3)  $\frac{3h}{2\pi}$

(4)  $\frac{4h}{2\pi}$

55. A radioactive sample of half life 10 days contains  $1000 \times$  nuclei. Number of original nuclei present after 5days is

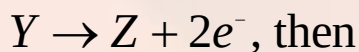
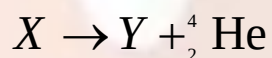
(1)  $707x$

(2)  $750x$

(3)  $500x$

(4)  $250x$

56. An element  $X$  decays into element  $Z$  by two step process



(1)  $X$  and  $Z$  are isobars

(2)  $X$  and  $Y$  are isotopes

(3)  $X$  and  $Z$  are isotones

(4)  $X$  and  $Z$  are isotopes

57. A nucleus of mass 20 u emits a  $\gamma$ -photon of energy 6 MeV. If the emission assume to occur when nucleus is free and rest, and then the nucleus will have kinetic energy nearest to (Take  $1 \text{ u} = 1.6 \times 10^{-27} \text{ kg}$ )

- (1) 10 keV
- (2) 1 keV
- (3) 0.1 keV
- (4) 100 keV

58. Constant DC voltage required from a variable AC voltage. Which of the following is correct order of operation?

- (1) Regulator, filter, rectifier
- (2) Rectifier, regulator, filter
- (3) Rectifier, filter, regulator
- (4) Filter, regulator, rectifier

59. In a transistor, the collector current varies by 0.49 mA and emitter current varies by 0.50 mA, Current gain  $\beta$  measured is

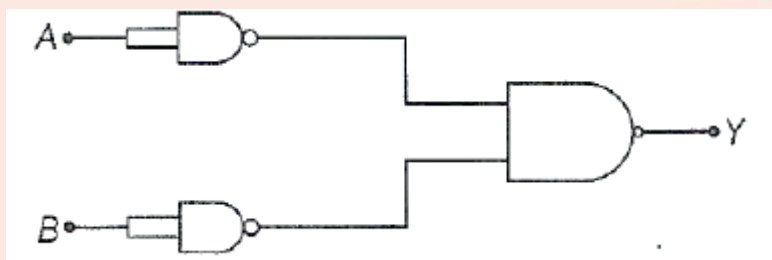
- (1) 49

(2) 150

(3) 99

(4) 100

60. Identify the logical operation carried out by the following circuit



(1) AND

(2) NAND

(3) NOR

(4) OR

**KCET-2016-Set-A-1**  
**Answer keys**

|    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|
| 1  | 3 | 16 | 1 | 31 | 1 | 46 | 3 |
| 2  | 2 | 17 | 3 | 32 | 2 | 47 | 1 |
| 3  | 1 | 18 | 3 | 33 | 1 | 48 | 4 |
| 4  | 2 | 19 | 1 | 34 | 2 | 49 | 2 |
| 5  | 1 | 20 | 4 | 35 | 3 | 50 | 3 |
| 6  | 2 | 21 | 1 | 36 | 4 | 51 | 1 |
| 7  | 3 | 22 | 3 | 37 | 1 | 52 | 4 |
| 8  | 4 | 23 | 1 | 38 | 2 | 53 | 4 |
| 9  | 2 | 24 | 4 | 39 | 3 | 54 | 3 |
| 10 | 1 | 25 | 2 | 40 | 1 | 55 | 4 |
| 11 | 2 | 26 | 4 | 41 | 4 | 56 | 4 |
| 12 | 4 | 27 | 1 | 42 | 2 | 57 | 2 |
| 13 | 2 | 28 | 3 | 43 | 3 | 58 | 3 |
| 14 | 3 | 29 | 4 | 44 | 4 | 59 | 1 |
| 15 | 4 | 30 | 3 | 45 | 1 | 60 | 4 |

**KCET-2016-Set-A-1****Solutions – Physics**

1. The total distance covered by the body in  $t$  seconds is,

$$s = \frac{1}{2}gt^2$$

Substitute  $10 \text{ ms}^{-2}$  for  $g$  in  $10 \text{ s}$  for  $t$  then,

$$\begin{aligned}s &= \frac{1}{2}(10 \text{ ms}^{-2})(10 \text{ s})^2 \\ &= 500 \text{ m}\end{aligned}$$

The average velocity is given by,

$$\begin{aligned}v_{\text{av}} &= \frac{s}{t} \\ &= \frac{500 \text{ m}}{10 \text{ s}} \\ &= 50 \text{ ms}^{-1}\end{aligned}$$

2. The range is maximum at the angle of  $\theta = 45^\circ$ . The range of the complementary angles is equal.

The range of the two angles of projection  $\theta_1$  and  $\theta_2$  is same if their sum is  $90^\circ$ . So, the range of A and C is same and will be less than that of B because it has angle  $\theta = 45^\circ$ .

Therefore,  $R_A = R_C < R_B$ .

3. The component of  $r$  along x-axis is given by,

$$r = E \cos \theta$$

The above component is maximum when  $\theta = 0$ .

Therefore, for maximum value  $r$  is along positive x-axis.

4. For the motion, the maximum static frictional force should be less than or equal to the force  $F = ma$ . That is

$$ma = \mu_s mg$$

Use  $\mu_s = 0.3$ ,  $g = 10 \text{ ms}^{-2}$  in the above relation then,

$$ma = (0.3)m(10 \text{ ms}^{-2})$$

$$a = 3 \text{ ms}^{-2}$$

5. The momentum of the system remains conserved in explosion.

The mass is  $m = 12 \text{ kg}$  and the initial velocity is  $u = 0$ .

The initial momentum is given by,

$$m \times u = 0$$

The momentum of the piece of 4 kg is given by,

$$4 \times v = 20$$

Consider above,

$$\begin{aligned} v &= \frac{20}{4} \\ &= 5 \text{ ms}^{-1} \end{aligned}$$

Apply the conservation of the momentum before and after explosion gives,

$$m_1 v + m_2 v' = 0$$

$$20 + 8 \times v' = 0$$

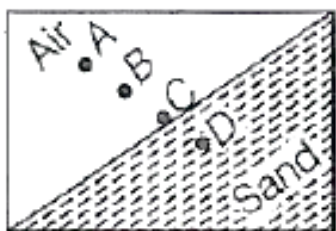
$$\begin{aligned} v' &= \frac{-20}{8} \\ &= \frac{-5}{2} \text{ ms}^{-1} \end{aligned}$$

The kinetic energy after explosion is given by,

$$\begin{aligned} \frac{1}{2} m v'^2 &= \frac{1}{2} \times 8 \times \left( \frac{5}{2} \right)^2 \\ &= \frac{1}{2} \times 8 \times \frac{25}{4} \\ &= 25 \text{ J} \end{aligned}$$

6. Consider the diagram,





Since, the centre of mass lies towards the heavier mass, from the above diagram the mass is more at the point D. so centre of mass should be at D.

7. Let the velocity of the ring R be  $v_R$ , the velocity of cylinder C be  $v_C$  and the velocity of sphere S be  $v_S$ .

The velocity of the rolling without slipping down the inclined plane is given by,

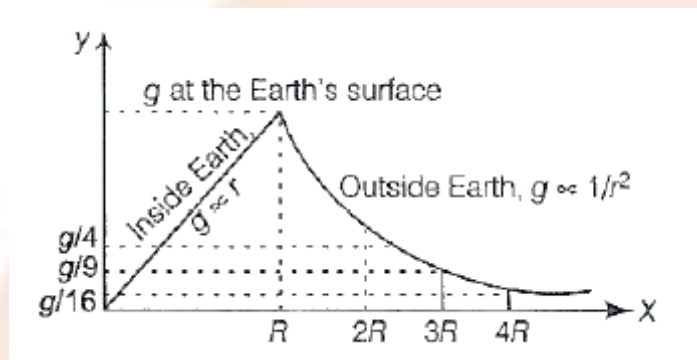
$$v = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}}$$

Since, the  $\frac{k^2}{R^2}$  is minimum in case of sphere and maximum in case of ring. So, the velocity will be maximum for sphere and minimum for ring.

Therefore,  $v_S > v_C > v_R$  or  $v_R < v_C < v_S$ .

8. The value of  $g$  is 0 at the center of the earth and maximum at the surface. The value of  $g$  starts to decrease in one goes away from the surface of the earth.

Therefore, the graph (d), is showing the correct variation of  $g$ .



9. When the spring is stretched then the strain produced in the spring is shear strain.
10. The continuity equation is given by,

$$a_1 v_1 = a_2 v_2$$

$$\frac{v_1}{v_2} = \frac{a_2}{a_1}$$

The area of the pipe is given by,

$$a = \frac{\pi}{4} \left( \frac{d}{2} \right)^2$$

This results in,

$$\begin{aligned}\frac{v_1}{v_2} &= \frac{\pi d_2^2/8}{\pi d_1^2/8} \\ &= \frac{d_2^2}{d_1^2}\end{aligned}$$

Here  $d_1 = 5$  cm, and  $d_2 = 10$  cm then,

$$\begin{aligned}\frac{v_1}{v_2} &= \left(\frac{10 \text{ cm}}{5 \text{ cm}}\right)^2 \\ &= \frac{100}{25} \\ &= \frac{4}{1} \text{ or } 4:1\end{aligned}$$

11. For the first case,

The average temperature is given by,

$$\begin{aligned}\frac{94 + 86}{2} &= \frac{180}{2} \\ &= 90^\circ\text{C}\end{aligned}$$

So, the excess temperature is,

$$90^\circ - 20^\circ = 70^\circ\text{C}$$

The rate of fall of the temperature is given by,

$$\begin{aligned}\frac{d\theta_1}{dt} &= -\left[\frac{94-86}{2}\right] \\ &= -\frac{8}{2} \\ &= -4^\circ\text{C min}^{-1}\end{aligned}$$

From Newton's law of cooling,

$$\frac{d\theta}{dt} \propto (\theta - \theta_0) = -k(\theta - \theta_0)$$

This implies,

$$-4 = k \times 70 \quad \dots\dots (I)$$

For the second case,

The average temperature is given by,

$$\begin{aligned}\frac{74+66}{2} &= \frac{140}{2} \\ &= 70^\circ\text{C}\end{aligned}$$

So, the excess temperature is,

$$70^\circ - 20^\circ = 50^\circ\text{C}$$

The rate of fall of the temperature is given by,

$$\begin{aligned}\frac{d\theta_1}{dt} &= -\left[\frac{74-66}{t_2}\right] \\ &= -\frac{8^\circ\text{C}}{t_2} \text{ min}^{-1}\end{aligned}$$

This implies,

$$-\frac{8^{\circ}\text{C}}{t_2} = k \times 50 \quad \dots\dots (II)$$

Divide equation (I) and (II).

$$\begin{aligned} \frac{t_2}{2} &= \frac{7}{5} \\ t_2 &= \frac{14}{5} \\ &= 2.8 \text{ min} \end{aligned}$$

12. The rate of the flow of heat has relation,

$$\frac{dQ}{dt} \propto \frac{A\Delta T}{l}$$

Consider above relation, the option  $r = 2 \text{ m}$  and  $l = \frac{1}{2} \text{ m}$  will conduct most heat.

13. The expression for the efficiency of the Carnot engine is,

$$\begin{aligned} \eta &= \frac{T_1 - T_2}{T_1} \\ &= \frac{W}{Q_1} \end{aligned}$$

Substitute  $W = 800 \text{ J}$ ,  $T_1 = 400 \text{ K}$  and  $T_2 = 300 \text{ K}$  to obtain,

$$\frac{800}{Q_1} = \frac{400 - 300}{400}$$

$$Q_1 = \frac{400}{100} \times 800$$

$$= 3200 \text{ J}$$

14. The maximum value in the SHM is given by,

$$v_{\max} = \omega A \quad \dots\dots (I)$$

The maximum acceleration is given by,

$$\alpha_{\max} = A\omega^2 \quad \dots\dots (II)$$

Divide equation (I) and (II).

$$\frac{\alpha_{\max}}{v_{\max}} = \frac{A\omega^2}{\omega A}$$

$$= \omega$$

Substitute  $v_{\max} = 0.5 \text{ ms}^{-1}$  and  $\alpha_{\max} = 1.0 \text{ ms}^{-2}$ .

$$\omega = \frac{1}{0.5}$$

$$= 2 \text{ rad s}^{-1}$$

15. Consider Doppler equation is the first case,

$$f' = \frac{v}{v - v_s} f$$

Substitute  $v = 350 \text{ ms}^{-1}$  and  $v_s = 50 \text{ ms}^{-1}$ .

$$f' = \frac{350}{350 - 50} f \quad \dots\dots (I)$$

Consider Doppler equation in the second case.

$$\begin{aligned} f'' &= \frac{v}{v + v_s} f \\ &= \frac{350}{350 + 50} f \quad \dots\dots (II) \end{aligned}$$

From equation (I) and equation (II),

$$\begin{aligned} \frac{f''}{500} &= \frac{350/400}{350/300} \\ &= \frac{3}{4} \\ f'' &= \frac{3}{4} \times 500 \\ &= 375 \text{ Hz} \end{aligned}$$

16. From the gauss law of magnetism, if there is only one charge in the universe then,

$\oint E \cdot dS = 0$ , if charge is outside

$$= \frac{q}{\epsilon_0}, \text{ if charge is inside}$$

17. The total force on the electron is given by,

$$F = eE$$

The force due to motion is given by,

$$F = ma$$

This implies,

$$eE = ma$$

$$a = \frac{eE}{m}$$

The distance travelled by a fallen body is given by,

$$h = ut + \frac{1}{2}at^2$$

Substitute 0 for  $u$  and  $\frac{eE}{m}$  for  $a$  then,

$$h = (0)t + \frac{1}{2}\left(\frac{eE}{m}\right)t^2$$

$$= \left(\frac{eE}{m}\right)t^2$$

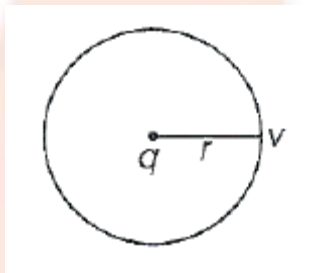
$$t = \sqrt{\frac{2hm}{eE}}$$



18. The field at the axial point of the dipole is twice the field at equatorial point. This implies,

$$E_{\text{axial}} = -2E_{\text{equatorial}}$$

19. On an equipotential surface all the point are at same distance from the charge. This is possible in the case shown below,



Therefore, the equipotential surface is a sphere with charge at center.

20. From the conservation of the energy,

$$\begin{aligned}
\frac{Kq_1q_2}{r_1} + \frac{1}{2}mu^2 &= \frac{Kq_1q_2}{r_2} + \frac{1}{2}mv^2 \\
\frac{Kq_1q_2}{r_1} &= \frac{Kq_1q_2}{r_2} + \frac{1}{2}mv^2 \\
\frac{1}{2}mv^2 &= Kq_1q_2 \left[ \frac{1}{r_1} - \frac{1}{r_2} \right] \\
v^2 &= \frac{2Kq_1q_2(r_2 - r_1)}{mr_1r_2} \quad \dots\dots (I)
\end{aligned}$$

The fixed charges is,

$$\begin{aligned}
q_1 &= 2 \text{ mC} \\
&= 2 \times 10^{-3} \text{ C}
\end{aligned}$$

The mass of the particle is,

$$\begin{aligned}
m &= 1 \text{ g} \\
&= 1 \times 10^{-3} \text{ kg}
\end{aligned}$$

The charge on the particle is,

$$\begin{aligned}
q_2 &= 1 \mu\text{C} \\
&= 1 \times 10^{-6} \text{ C}
\end{aligned}$$

The distance between the charges is,

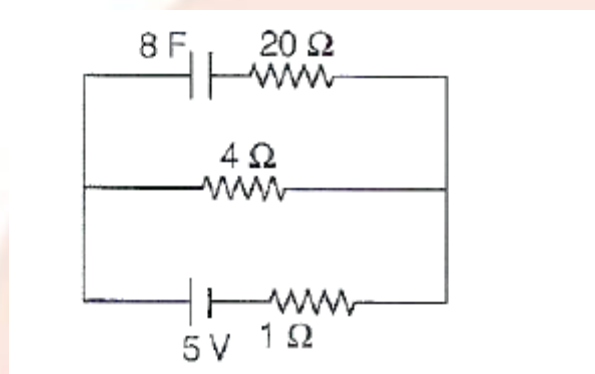
$$r_1 = 1 \text{ m}$$

Substitute these values in equation (I).

$$v^2 = \frac{2 \times 9 \times 10^9 \times 10^{-6} \times 2 \times 10^{-3} \times (10 - 1)}{10^{-3} \times 1 \times 10}$$

$$v = 10 \text{ ms}^{-1}$$

21. Consider the diagram,



The current in the zero through capacitor when it is fully charged.

The current is given by,

$$\begin{aligned} I &= \frac{V}{R_{eq}} \\ &= \frac{5}{1 + 4} \\ &= 1 \text{ A} \end{aligned}$$

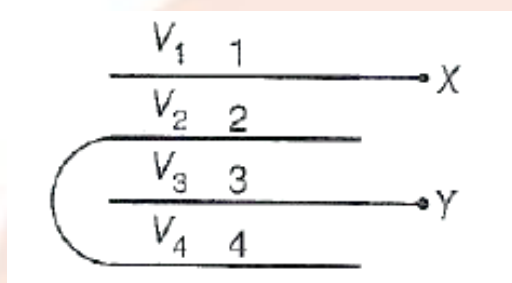
The potential difference across the capacitor is given by,

$$\begin{aligned} V &= 5 - 1 \times 1 \\ &= 4 \text{ V} \end{aligned}$$

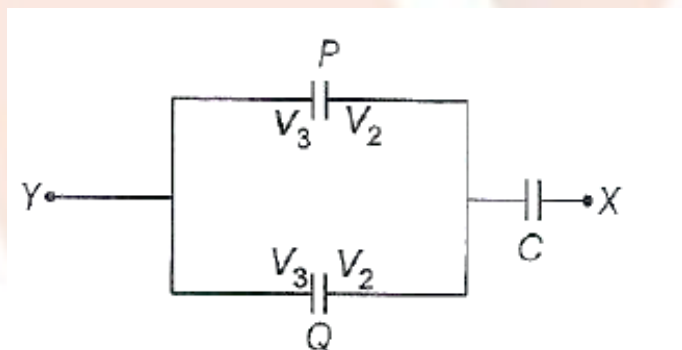
The charge stored in the capacitor is given by,

$$\begin{aligned}
 q &= CV \\
 &= 8 \times 4 \\
 &= 32 \text{ C}
 \end{aligned}$$

22. Consider the diagram,



From above figure it is clear that two capacitances forms the equivalent arrangement as shown below,



If  $C$  is the capacitance of each capacitor, then,

The net capacitance for  $P$  and  $Q$  in parallel is,

$$C_{PQ} = 2C$$

The  $C_{PQ}$  is in series with  $C$  so,

$$\begin{aligned}C_{XY} &= \frac{C \cdot C_{PQ}}{C + C_{PQ}} T \\&= \frac{C \cdot 2C}{C + 2C} \\&= \frac{2}{3} C\end{aligned}$$

The normal capacitance of parallel plate capacitor is given by,

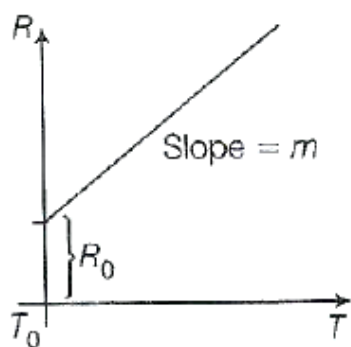
$$C = \frac{\epsilon_0 A}{d}$$

Thus,

$$C_{XY} = \frac{2}{3} \frac{\epsilon_0 A}{d}$$

23. The mobility of the free electrons in a conductor is directly proportional to the drift velocity which is directly proportional to relaxation time. This implies mobility is directly proportional to the relaxation time.

24. Consider the diagram,



The expression for resistance variation with temperature is given by,

$$R_t = R_0 (1 + \alpha \Delta t)$$

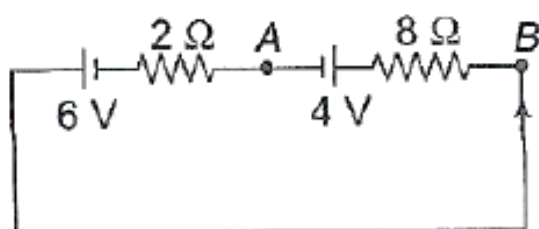
The slope of the graph is given by,

$$m = R_0 \alpha$$

So,

$$\alpha = \frac{m}{R_0}$$

25. Consider the circuit,



Apply kirchhoff's second law from A to B in anticlockwise direction.

$$I(2 + 8) = 6 - 4$$

$$I \cdot 10 = 2$$

$$I = 0.2 \text{ A}$$

Now,

$$V_A - I \times 2 + 6 = V_B$$

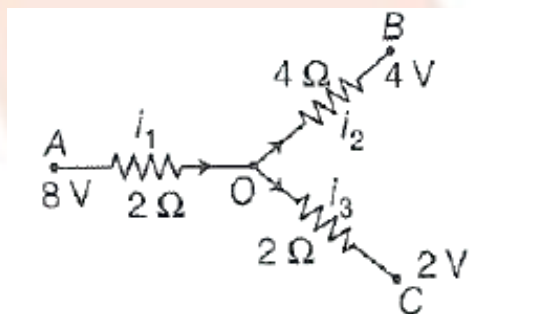
$$\begin{aligned} V_A - V_B &= 0.2 \times 2 - 6 \\ &= -5.6 \text{ V} \end{aligned}$$

The negative value implies,

$$V_A < V_B$$

Therefore,  $V_{BA} = 5.6 \text{ V}$

26. Consider the diagram,



Let the potential at O is  $V_0$ .

Then, apply kirchhoff's first law at O,

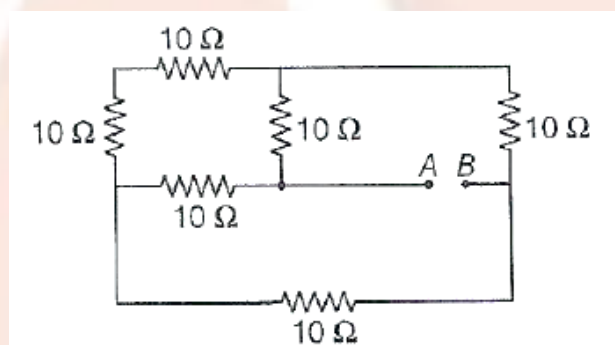
$$\frac{8 - V_0}{2} = \frac{V_0 - 4}{4} + \frac{V_0 - 2}{2}$$

$$= \frac{V_0 - 4 + 2V_0 - 4}{4}$$

$$16 - 2V_0 = 3V_0 - 8$$

$$V_0 = 4.8 \text{ V}$$

27. Consider the circuit,



The above circuit represents a balanced Wheatstone Bridge with each resistance equal to  $10 \Omega$ .

Therefore, the equivalent resistance is  $R_{eq} = 10 \Omega$ .

28. The expression for the power consumed is given by,

$$P = \frac{V^2}{R}$$

The ratio of the power consumed is given by,



$$\frac{P_1}{P_2} = \frac{\frac{V^2}{R_1}}{\frac{V^2}{R_2}}$$

$$= \frac{R_2}{R_1}$$

Substitute  $R_1 = 10 \, \Omega$  and  $R_2 = 20 \, \Omega$ .

$$\frac{P_1}{P_2} = \frac{20}{10}$$

$$= \frac{2}{1} \text{ or } 2:1$$

29. The force on the charged particle in a magnetic field is given by,

$$F = qvB \sin \theta$$

When  $\theta = 0$  along the x-axis,

$$F = qvB \sin(0)$$

$$= 0$$

This implies, the proton will continue to move with velocity  $v$  along the x-axis.

30. The expression for the velocity in the cyclotron is given by,

$$v = \frac{qBR}{m}$$

And,

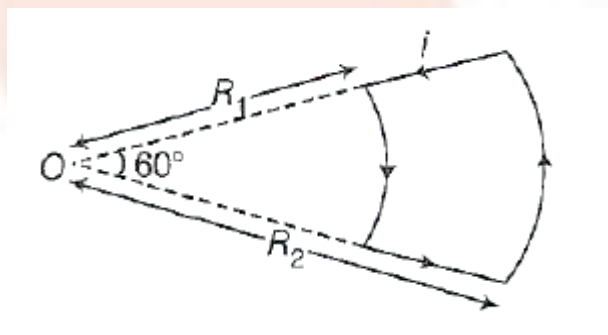
$$\omega = \frac{qB}{m}$$

Therefore, velocity increase with radius and  $\omega$  remains constant with increase of radius.

31. The expression for magnetic field at center due to an current carrying conductor is given by,

$$B = \frac{\mu_0}{4\pi} \left( \frac{i}{R} \right) \theta$$

Consider the diagram,



From above the magnetic field due to  $R_1$  is given by,

$$\begin{aligned} B_1 &= \frac{\mu_0}{4\pi} \left( \frac{i}{R_1} \right) 60^\circ \\ &= \frac{\mu_0}{4\pi} \left( \frac{i}{R_1} \right) \frac{\pi}{3} \end{aligned}$$

The magnetic field due to  $R_2$  is given by,

$$B_2 = \frac{\mu_0}{4\pi} \left( \frac{i}{R_2} \right) \frac{\pi}{3}$$

The net magnetic field is given by,

$$\begin{aligned} B &= B_1 - B_2 \\ &= \frac{\mu_0}{4\pi} \left( \frac{i}{R_1} \right) \frac{\pi}{3} - \frac{\mu_0}{4\pi} \left( \frac{i}{R_2} \right) \frac{\pi}{3} \\ &= \frac{\mu_0 i \pi}{4\pi \cdot 3} \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \\ &= \frac{\mu_0 i}{12} \left( \frac{1}{R_1} - \frac{1}{R_2} \right) \end{aligned}$$

32. The expression for change in charge from time 0 to 90 minutes is given by,

$$\begin{aligned}dQ &= \int_0^{90} Idt \\&= \int_0^{90} 20dt \\&= [20t]_0^{5400\text{ s}} \\&= 20 \times 5400\end{aligned}$$

Further simplify the above,

$$\begin{aligned}dQ &= 108000\text{ C} \\&= 10.8 \times 10^4\text{ C}\end{aligned}$$

33. The expression for high resistance in series in a Galvanometer is given by,

$$\begin{aligned}R &= \frac{V}{I_g} - G \\&= \left( \frac{20}{5 \times 10^{-3}} \right) - 50 \\&= 4000 - 50 \\&= 3950\ \Omega\end{aligned}$$

34. The relation of susceptibility with temperature is given by,

$$\begin{aligned}\chi &\propto \frac{1}{T} \\ \chi T &= \text{constant}\end{aligned}$$

This implies,

$$\chi_1 T_1 = \chi_2 T_2$$

35. The expression for magnetic field at a location is given by,

$$B = \frac{H}{\cos \delta}$$

Substitute 3.0 gauss for  $H$  and  $30^\circ$  for  $\delta$ .

$$\begin{aligned} B &= \frac{3.0}{\cos 30^\circ} \\ &= \frac{3.0}{\sqrt{3}/2} \\ &= \frac{2 \times 3}{\sqrt{3}} \\ &= 3.5 \text{ G} \end{aligned}$$

36. The process of superimposition of the message signal on the high frequency carrier wave is called modulation.

37. The expression for magnetic field stored per unit volume in a solenoid is given by,

$$U = \frac{\mu_0 n^2 I^2}{2}$$

Substitute  $n = 40 \text{ turns cm}^{-1}$  or  $40 \times 10^2 \text{ turns m}^{-1}$  and  $I = 1.00 \text{ A}$ .

$$U = \frac{4\pi \times 10^{-7} (40 \times 10^2)^2 (1.00 \text{ A})^2}{2}$$

$$= 3.2\pi \text{ Jm}^{-3}$$

38. From the Faraday's law, the induced EMF is given by,

$$E = \frac{d}{dt} \phi$$

$$= B \frac{dA}{dt}$$

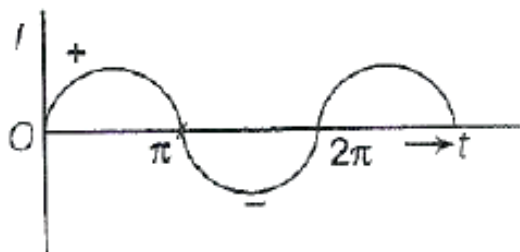
$$= B \cdot \frac{d}{dt} \frac{\pi R^2 \theta}{2\pi}$$

$$= \frac{1}{2} BR^2 \omega$$

In case of wheel  $R = L$  thus,

$$E = \frac{1}{2} BL^2 \omega$$

39. Consider the diagram,



The average value of the AC current over a cycle is always zero.

40. The capacitance of the capacitor is,

$$\begin{aligned}C &= 10 \mu\text{F} \\ &= 10 \times 10^{-6} \text{ F}\end{aligned}$$

The variation of the source voltage is,

$$V = 50\sqrt{2} \sin 100t$$

The current is given by,

$$I = \frac{V_{\text{rms}}}{1/\omega C}$$

The rms voltage is given by,

$$\begin{aligned}V_{\text{rms}} &= \frac{V}{\sqrt{2}} \\ &= \frac{50\sqrt{2}}{\sqrt{2}} \\ &= 50 \text{ V}\end{aligned}$$

The reading of the ammeter is,

$$\begin{aligned}50 \times 100 \times 10 \times 10^{-6} &= 50 \times 10^{-3} \text{ A} \\ &= 50 \text{ mA}\end{aligned}$$

41. The source voltage in a series LCR circuit is given by,

$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

Substitute  $V_R = 60 \text{ V}$ ,  $V_L = 40 \text{ V}$  and  $V_C = 120 \text{ V}$ .

$$\begin{aligned} V &= \sqrt{(60)^2 + (40 - 120)^2} \\ &= \sqrt{(60)^2 + (-80)^2} \\ &= \sqrt{3600 + 6400} \\ &= 100 \text{ V} \end{aligned}$$

42. The power dissipated in AC circuit is given by,

$$P = \frac{V_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}} \times \cos \phi$$

When  $\phi = 60^\circ$ ,

$$\begin{aligned} P &= \frac{V_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}} \times \cos 60^\circ \\ &= \frac{V_0 I_0}{\sqrt{2}} \frac{1}{2} \\ &= \frac{V_0 I_0}{4} \end{aligned}$$

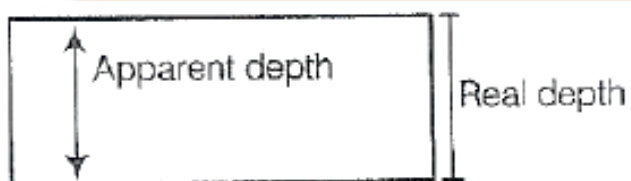
43. The electromagnetic radiation used to sterilize the milk in dairy is Ultraviolet or UV rays.



44. The expression for the apparent depth is given by,

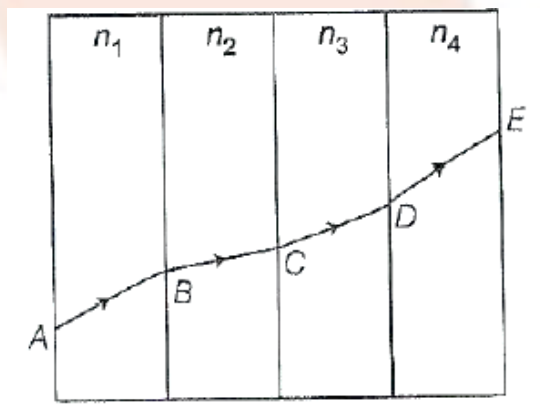
$$\text{Apparent depth} = \frac{\text{Real depth}}{\mu}$$

The refractive index  $\mu$  is inversely proportional to the wavelength.



The wavelength is least for violet so, refractive index is maximum for it and apparent depth will be minimum. That means the violet letter appears to be raised more.

45. Consider the diagram,



The rays are parallel in medium (1) and (4), this implies,

$$n_1 = n_4$$

46. The focal length of the lens is 20 cm and the refractive index is 1.5.

The image formed by a convex lens is erect and enlarged if object is placed between focus and optical center.

Therefore, the distance should be 15 cm.

47. The refractive index for the prism in position of minimum deviation is given by,

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\frac{A}{2}}$$

The prism angle is  $A = 60^\circ$  and the refractive index is  $\mu = \sqrt{2}$ .

Thus,

$$\sqrt{2} = \frac{\sin\left(\frac{60^\circ + \delta_m}{2}\right)}{\sin \frac{60^\circ}{2}}$$

$$\sin\left(\frac{60^\circ + \delta_m}{2}\right) = \frac{\sqrt{2}}{2}$$

$$\begin{aligned}\frac{60^\circ + \delta_m}{2} &= \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) \\ &= 45^\circ\end{aligned}$$

Further simplify the above,

$$\delta_m = 30^\circ$$

48. The alternate red and blue fringes produce incoherence that results in no interference.

49. The linear width of the central maxima is given by,

$$\text{Angular width} = \frac{2\lambda}{d}$$

The wavelength is,

$$\lambda = 600 \text{ nm}$$

$$= 600 \times 10^{-9} \text{ m}$$

The slit width is,

$$\begin{aligned}d &= 0.2 \text{ mm} \\ &= 0.2 \times 10^{-3} \text{ m}\end{aligned}$$

Thus, angular width is given by,

$$\begin{aligned}\text{Angular width} &= \frac{2(600 \times 10^{-9} \text{ m})}{0.2 \times 10^{-3} \text{ m}} \\ &= 6 \times 10^{-7} \times 10^4 \\ &= 6 \times 10^{-3} \text{ rad}\end{aligned}$$

50. The expression for Fresnel distance is,

$$D_f = \frac{a^2}{\lambda}$$

The aperture is,

$$\begin{aligned}a &= 4 \text{ mm} \\ &= 4 \times 10^{-3} \text{ m}\end{aligned}$$

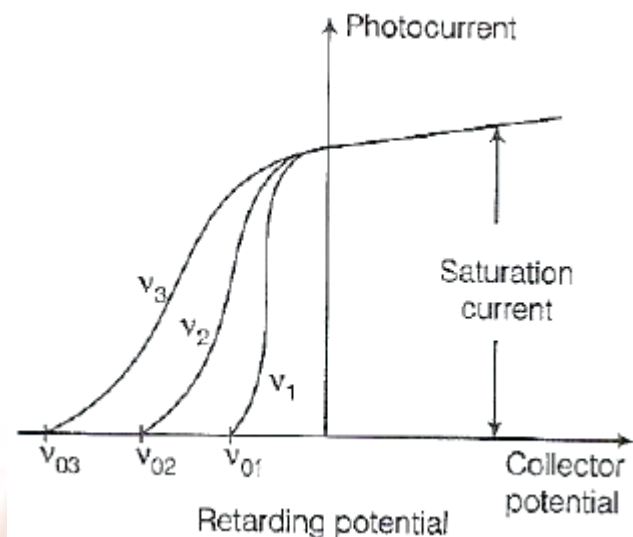
The wavelength is,

$$\begin{aligned}\lambda &= 400 \text{ nm} \\ &= 400 \times 10^{-9} \text{ m}\end{aligned}$$

So, the Fresnel distance is,

$$\begin{aligned}D_f &= \frac{(4 \times 10^{-3} \text{ m})^2}{400 \times 10^{-9} \text{ m}} \\ &= 40 \text{ m}\end{aligned}$$

51. Consider the diagram,



The smaller is the frequency of incidence, the smaller is the stopping voltage required.

This implies,

$$V_3 > V_2 > V_1$$

52. The expression for de-Broglie wavelength is given by,

$$\lambda = \frac{12.3}{\sqrt{V}} \text{ \AA}$$

Substitute 400 V for  $V$ .

$$\begin{aligned}\lambda &= \frac{12.3}{\sqrt{400}} \text{ \AA} \\ &= \frac{12.3}{20} \text{ \AA} \\ &= 0.615 \text{ \AA} \\ &= 0.0615 \text{ nm}\end{aligned}$$

53. The total energy is the sum of potential energy and kinetic energy,

$$-3.4 \text{ eV} = \text{K.E} + \text{P.E} \quad \text{..... (I)}$$

The relation between potential and kinetic energy is given by,

$$\text{P.E} = -2\text{K.E} \quad \text{..... (II)}$$

Use the relation in equation (I),

$$-3.4 \text{ eV} = \text{K.E} - 2\text{K.E}$$

$$\text{K.E} = 3.4 \text{ eV}$$

From equation (II),

$$\begin{aligned}\text{P.E} &= -2(3.4 \text{ eV}) \\ &= -6.8 \text{ eV}\end{aligned}$$

54. From Bohr's postulate the angular momentum is given by,

$$mvr = \frac{nh}{2\pi}$$

For  $n = 4$ , the angular momentum is,

$$mvr_4 = \frac{4h}{2\pi} \quad \dots\dots (I)$$

For  $n = 1$ , the angular momentum is,

$$mvr_1 = \frac{1h}{2\pi} \quad \dots\dots (II)$$

Thus, the change in angular momentum from equation (I) and (II) is,

$$\begin{aligned} \Delta mvr &= \frac{4h}{2\pi} - \frac{h}{2\pi} \\ &= \frac{3h}{2\pi} \end{aligned}$$

55. The expression for nuclei left after half life decay is given by,

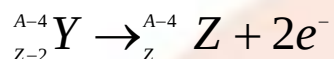
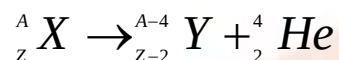
$$N = \frac{N_0}{2^n}$$

If the half life is  $t_{\frac{1}{2}} = 5$  days then, life period will be  $t = 10$  days

For  $n = \frac{1}{2}$  and  $N_0 = 1000x$ .

$$\begin{aligned} N &= \frac{1000x}{2^{\frac{1}{2}}} \\ &= 707x \end{aligned}$$

56. Let  $A$  and  $Z$  be the mass number and atomic number of any element  $X$  respectively, then,



This implies  $X$  and  $Z$  are isotopes.

57. The expression for relation between kinetic energy and momentum is,

$$\text{K.E} = \frac{p^2}{2m} \quad \dots\dots (I)$$

The expression for relation between momentum and total energy is,

$$E = pc$$

$$p = \frac{E}{c}$$

Consider above relation equation (I) becomes,



$$\begin{aligned} \text{K.E} &= \frac{(E/c)^2}{2m} \\ &= \frac{(6 \times 10^6 \times 1.6 \times 10^{-19} / 3 \times 10^8)^2}{2(20 \times 1.6 \times 10^{-27})} \\ &= 1.6 \times 10^{-15} \text{ J} \\ &= \frac{1.6 \times 10^{-16}}{1.6 \times 10^{-16}} \text{ keV} \end{aligned}$$

Further simplify the above gives,

$$\text{K.E} = 1 \text{ keV}$$

58. A filter is used in the output circuit of a rectifier to smooth the output current.

Therefore, the rectifier, filter, regular is correct option.

59. The expression for current gain is given by,

$$\beta = \frac{\alpha}{1 - \alpha} \quad \dots\dots (I)$$

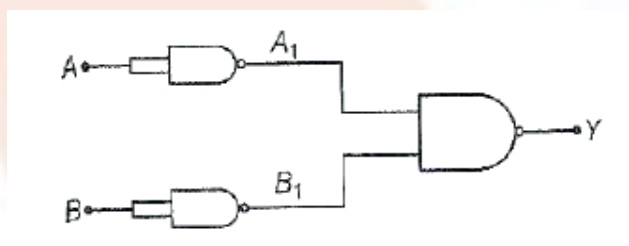
The expression to calculate  $\alpha$  is given by,

$$\begin{aligned}\alpha &= \frac{\Delta I_C}{\Delta I_E} \\ &= \frac{0.49 \text{ mA}}{0.50 \text{ mA}} \\ &= 0.98\end{aligned}$$

Substitute 0.98 for  $\alpha$  in equation (I).

$$\begin{aligned}\beta &= \frac{0.98}{1 - 0.98} \\ &= \frac{0.98}{0.02} \\ &= 49\end{aligned}$$

60. Consider the diagram,



The truth table for the above combination is given by,

| A | B | $A_1$ | $B_1$ | Y |
|---|---|-------|-------|---|
| 0 | 0 | 1     | 1     | 0 |
| 0 | 1 | 1     | 0     | 1 |
| 1 | 0 | 0     | 1     | 1 |
| 1 | 1 | 0     | 0     | 1 |